

Pratik Kumar Padhi

+91 7894139410 | pratikpadhi00@gmail.com | linkedin.com/in/pratikpadhi | github.com/pratikpadhi | Bhubaneswar, India

SUMMARY

AI/ML engineering student specializing in computer vision and agricultural AI, with hands-on experience building and deploying end-to-end deep learning pipelines. Conference presenter at ICCAIML 2026 with co-authored research on AI-driven crop health monitoring. Proficient in Python, PyTorch, TensorFlow, and AWS deployment. Seeking internship roles in AI/ML, Data Science, or Software Engineering to contribute research depth and production-level project experience.

EDUCATION

Centurion University of Technology and Management

B.Tech in Computer Science and Engineering (AI & Machine Learning Specialization)

Bhubaneswar, Odisha

Aug 2023 – May 2027

- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Computer Vision, Statistics for Data Science

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript

ML/AI Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLOv8, Hugging Face

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Cloud & DevOps: AWS (EC2, S3, Lambda), Git, GitHub, Linux, Docker (basic)

Web: HTML, CSS, React (basic), Flask/FastAPI, Netlify

PROJECTS

Cropsensy — AI-Powered Crop Prediction Platform | *Python, TensorFlow, AWS, Flask* | [Live Demo](#) 2024 – Present

- Engineered an end-to-end crop yield prediction pipeline using a CNN-LSTM hybrid model trained on multi-variable soil, weather, and environmental datasets, achieving **92% classification accuracy** on held-out test data.
- Deployed the full-stack application on **AWS EC2 + S3** with a Flask REST API backend, achieving **99.5% uptime** over a 6-month production window.
- Implemented automated data preprocessing and feature engineering pipeline reducing model retraining time by **30%**, enabling weekly model updates with fresh sensor data.

Real-Time Weed Detection System | *YOLOv8, PyTorch, OpenCV, Python*

Jan 2025 – Apr 2025

- Built a real-time agricultural weed classification system using **YOLOv8** fine-tuned on a custom-annotated dataset of 12 weed species, achieving **94% mAP@0.5** on the validation set.
- Optimized inference pipeline with TorchScript model export and batch processing, reducing per-frame latency from 120ms to **72ms** (40% improvement), enabling real-time field deployment on edge hardware.
- Presented findings at **ICCAIML 2026** and co-authored a research paper demonstrating 25% improvement in detection reliability over traditional threshold-based methods.

Coral Reef Health Classification System | *TensorFlow, Keras, OpenCV, Python*

Aug 2024 – Dec 2024

- Developed a 3-class image classification model (healthy / bleached / degraded) using a **ResNet-50 transfer learning** architecture fine-tuned on 8,000 labeled underwater images, achieving **94% test accuracy**.
- Built an automated monitoring dashboard with real-time image ingestion and anomaly alerting, reducing manual coral assessment review time by **70%** in pilot testing.
- Optimized image preprocessing pipeline using **OpenCV** and NumPy vectorization, achieving **12× throughput increase** over the baseline sequential processing approach.

RESEARCH & CERTIFICATIONS

ICCAIML 2026 — International Conference on AI and Machine Learning

Presenter & Co-Author: “Deep Learning for Agricultural Automation”

2026

- Co-authored peer-reviewed research demonstrating **25% improvement** in crop health monitoring accuracy using deep learning over traditional image processing baselines.

Java Programming — GeeksforGeeks (2026) **Target:** AWS Cloud Practitioner (in progress)